CS ANSWERSS FOR GTU

1. What is Information Security?

Information Security is the practice of protecting information and systems from unauthorized access, misuse, or damage. It ensures that data remains safe throughout its lifecycle—whether it is stored, processed, or transmitted.

The main goal is to preserve three core principles: confidentiality (only authorized access), integrity (data remains accurate and unchanged), and availability (data is accessible when needed). These principles form the foundation of all security measures.

2. List Types of hackers

Types of hackers are generally classified based on their intent, methods, and ethical stance toward systems they access.

A **White Hat hacker** is an ethical security professional who uses their skills to identify and fix vulnerabilities. They operate with permission and are essential to strengthening cybersecurity systems.

A **Black Hat hacker**, in contrast, acts with malicious intent. These individuals exploit vulnerabilities to steal data, cause damage, or gain unauthorized benefits, often for financial or personal motives.

A **Grey Hat hacker** exists between these two extremes. They may access systems without permission but typically do not intend harm; however, their actions still violate ethical and legal boundaries.

3. What is the default username and password for kali Linux?

The default username and password for **Kali Linux** are traditionally set for ease of initial access in a controlled environment.

The default username is **kali**, and the default password is also **kali**.

However, in modern versions of Kali Linux, especially after 2020, this default remains the same only for live environments or standard installations, but users are often encouraged—or required—to set their own credentials during installation for better security.

4. Describe CIA triad with example.

The CIA Triad is a fundamental model in Information Security that represents three essential principles: confidentiality, integrity, and availability.

Confidentiality ensures that sensitive information is accessible only to authorized individuals. For example, a password-protected bank account prevents unauthorized users from viewing personal financial data.

Integrity refers to maintaining the accuracy and consistency of data. For instance, when a student's grades are stored in a system, they must not be altered improperly or without authorization.

Availability ensures that information and systems are accessible when needed. An example would be an online banking system being available to users whenever they want to check their balance or perform transactions.

Example from online banking.

Consider a user accessing their bank account through a mobile app. **Confidentiality** is maintained because only the rightful user can log in using credentials like a password or OTP, ensuring no unauthorized person can view the account details. **Integrity** is preserved because the transaction data—such as account balance or transfer amount—cannot be altered improperly; if the user transfers ₹5,000, the system must record exactly ₹5,000, not more or less. **Availability** is ensured when the banking system remains accessible whenever the user needs it, allowing them to check balances or make transactions without downtime.

5. Explain CIA Triad in detail.

The CIA Triad, when explained in detail, forms the conceptual backbone of all Information Security systems, guiding how data is protected and managed.

Confidentiality focuses on restricting access to sensitive information strictly to authorized individuals. This is achieved through mechanisms such as encryption, authentication, and access control. For example, encrypted communication ensures that even if data is intercepted, it cannot be understood by unauthorized parties.

Integrity ensures that information remains accurate, consistent, and unaltered unless modified by authorized users in a legitimate manner. Techniques such as hashing, checksums, and digital signatures are used to detect and prevent unauthorized changes to data.

Availability guarantees that information and systems are accessible to authorized users whenever required. This involves maintaining system uptime, using backup systems, and protecting against attacks like denial-of-service that aim to disrupt access.

6. Explain MD5 hashing algorithm

MD5 (Message Digest Algorithm 5) is a widely known cryptographic hashing algorithm designed to convert input data of any length into a fixed-length 128-bit hash value.

Its primary purpose is to ensure data integrity. When data is passed through the MD5 algorithm, it produces a unique hash; even a small change in the input results in a completely different hash output. This property makes it useful for verifying whether data has been altered.

However, from a modern security perspective, MD5 is no longer considered secure for cryptographic purposes because it is vulnerable to collision attacks, where two different inputs can produce the same hash value.

7. Explain Md5 algorithm steps.

The MD5 algorithm operates through a sequence of well-defined steps that transform input data into a 128-bit hash value.

First, the input message is converted into a binary format and padded so that its length becomes 64 bits short of a multiple of 512 bits. Then, the original length of the message is appended to this padded data.

Next, the algorithm initializes four fixed 32-bit variables, commonly referred to as buffers. These serve as the starting state for the hashing process.

The padded message is then divided into 512-bit blocks. Each block undergoes four rounds of complex operations involving nonlinear functions, bitwise operations, and modular addition. These rounds progressively mix the input data with the buffer values.

Finally, after processing all blocks, the outputs of the buffers are combined to produce the final 128-bit hash value.

8. Explain SHA algorithm

The SHA (Secure Hash Algorithm) is a family of cryptographic hash functions designed to generate a fixed-size hash value from input data of any length, primarily used to ensure data integrity and security.

When data is processed through SHA, it produces a unique hash output; even a slight change in the input results in a completely different hash. This makes it highly effective for detecting data tampering.

Unlike MD5, modern versions such as SHA-256 are considered secure and are widely used in applications like digital signatures, SSL certificates, and blockchain systems.

9. Explain RSA algorithm

The RSA algorithm is a fundamental public-key cryptographic technique used for secure data transmission.

It is based on the mathematical difficulty of factoring large prime numbers. In RSA, two keys are generated: a public key, which is shared openly, and a private key, which is kept secret. Data encrypted with the public key can only be decrypted with the corresponding private key, ensuring secure communication.

The process begins with selecting two large prime numbers and computing their product. From this, a public key and private key are mathematically derived. When a sender wants to transmit secure information, they encrypt the message using the receiver’s public key, and only the receiver can decrypt it using their private key.

RSA is widely used in secure communications such as HTTPS, email encryption, and digital signatures.

10. List inventors of RSA. Write steps of RSA algorithm

The RSA algorithm was invented by Ron Rivest, Adi Shamir, and Leonard Adleman.

The working of RSA can be understood through its key generation and encryption–decryption process.

First, two large prime numbers are selected and multiplied to produce a modulus. Then, a value called Euler's totient is calculated. After that, a public exponent is chosen such that it is relatively prime to the totient. Using this, a private key is computed as the multiplicative inverse of the public exponent.

For encryption, the sender uses the receiver's public key to transform the plaintext into ciphertext using modular exponentiation. For decryption, the receiver applies their private key to the ciphertext to recover the original message.

This mathematical relationship ensures that even though the public key is known, the private key cannot be easily derived, which is what makes RSA secure.

11. Give comparison between Public key and Private Key cryptography.

Aspect	**Private Key Cryptography**	**Public Key Cryptography**
Keys Used	Single shared key	Two keys (public and private)
Encryption/Decryption	Same key used for both	Different keys used
Speed	Faster and efficient	Slower due to complex computations
Security	Less secure due to key sharing problem	More secure as no need to share private key
Key Distribution	Difficult and risky	Easier and more secure
Usage	Bulk data encryption	Secure key exchange, digital signatures
Example Algorithms	AES, DES	RSA, ECC

12. What is virus? Explain Virus Life cycle.

A computer virus is a type of malicious software that attaches itself to a legitimate program or file and spreads from one system to another when that program is executed. Its primary purpose is to disrupt operations, damage data, or gain unauthorized access.

The life cycle of a virus can be understood as a sequence of stages through which it operates. Initially, in the dormant phase, the virus remains inactive within the system. It then enters the propagation phase, where it replicates itself and spreads to other files or systems. In the triggering phase, a specific condition—such as a date or user action—activates the virus. Finally, in the execution phase, the virus performs its intended malicious activity, such as corrupting files or slowing down the system.

13. Define: Firewall. List limitations of firewall.

A firewall is a security system—either hardware, software, or a combination of both—that monitors and controls incoming and outgoing network traffic based on predefined security rules. Its primary purpose is to act as a barrier between a trusted internal network and untrusted external networks, such as the internet.

Despite its importance, a firewall has certain limitations. It cannot protect against attacks that bypass it, such as those originating from within the network.

It is also ineffective against social engineering attacks, where users are tricked into revealing sensitive information. Additionally, if malicious content is allowed through permitted ports or services, the firewall may not detect it.

Finally, improper configuration of firewall rules can weaken its effectiveness.

14. Define: Trojans, Rootkit, Backdoors, Keylogger

Trojans, rootkits, backdoors, and keyloggers are different types of malicious software, each with a specific purpose in compromising system security.

A Trojan (or Trojan horse) is a malicious program that disguises itself as legitimate software to trick users into installing it. Once activated, it can perform harmful actions such as stealing data or creating unauthorized access.

A rootkit is a type of malware designed to gain and maintain privileged access to a system while remaining hidden. It modifies system processes to avoid detection and allows attackers to control the system secretly.

A backdoor is a hidden method of bypassing normal authentication mechanisms, allowing unauthorized users to access a system remotely without being detected.

A keylogger is a program that records keystrokes made by a user, capturing sensitive information such as passwords, credit card numbers, and messages.

15. Define: Digital Signature. Also discuss various application areas of Digital Signature.

A digital signature is a cryptographic technique used to verify the authenticity and integrity of a message, document, or digital data. It ensures that the message truly comes from the claimed sender and has not been altered during transmission.

Digital signatures are widely used in various application areas.

In electronic commerce, they secure online transactions and agreements.

In email communication, they verify the identity of the sender and ensure message integrity.

In legal and governmental processes, digital signatures are used for signing official documents electronically.

They are also essential in software distribution, where they verify that software has not been tampered with and comes from a trusted source.

16. Explain step of creating and verifying digital signature.

In the signing phase, the sender first applies a hash function to the original message to generate a fixed-length message digest. This digest uniquely represents the content of the message. The sender then encrypts this digest using their private key. The result of this encryption is the digital signature, which is sent along with the original message.

In the verification phase, the receiver also applies the same hash function to the received message to generate a new message digest. Then, the receiver decrypts the digital signature using the sender's public key to obtain the original digest. If both digests match, it confirms that the message has not been altered and that it indeed came from the claimed sender.

17. Difference between Private key and Public Key cryptography

Same as 11 question

18. Define database forensics and list different kind of activities performed during database forensics.

Database forensics is a branch of digital forensics that focuses on the investigation of databases and related systems to uncover evidence of unauthorized activities, data breaches, or malicious actions. It involves analyzing database contents, metadata, logs, and transactions to determine what happened, how it happened, and who was responsible.

During database forensics, several key activities are performed. Investigators examine database logs to trace user actions and transactions. They analyze data modifications to identify unauthorized changes or deletions. Recovery of deleted or hidden data is also carried out. Additionally, access control mechanisms are reviewed to detect misuse of privileges, and timelines are reconstructed to understand the sequence of events.

19. Write advantages and disadvantages of firewall.

ADVANTAGES:

A firewall provides strong protection by filtering incoming and outgoing traffic based on predefined rules, thereby preventing unauthorized access to a network.

It helps in blocking malicious traffic

Reducing the risk of cyberattacks, and can also monitor and log network activity for security analysis.

it acts as a first line of defense between a trusted internal network and untrusted external networks.

DISADVANTAGES:

It cannot protect against internal threats that originate within the network.

It is ineffective against attacks that bypass it, such as those delivered through social engineering.

If malicious data passes through allowed ports or services, the firewall may fail to detect it.

Incorrect configuration can reduce its effectiveness and even create vulnerabilities.

20. List critical steps in preserving digital evidence.

Preserving digital evidence is a critical aspect of any forensic investigation, as improper handling can render the evidence invalid in legal proceedings.

The process begins with identifying and securing the evidence source to prevent any tampering. Once identified, the data must be carefully collected using forensically sound methods, ensuring that the original data is not altered. Creating an exact copy, often called imaging, is essential so that analysis is performed on the duplicate rather than the original.

Maintaining a proper chain of custody is equally important, documenting who handled the evidence, when, and for what purpose. Additionally, the integrity of the evidence must be verified using techniques like hashing to ensure it remains unchanged throughout the investigation.

21. Sketch IPsec Tunnel Mode and Transport mode.

IPsec (Internet Protocol Security) operates in two main modes: tunnel mode and transport mode, both used to secure IP communications but in different ways.

In transport mode, only the payload (the actual data being transmitted) of the IP packet is encrypted and/or authenticated, while the original IP header remains unchanged. This mode is typically used for end-to-end communication between two hosts, as it adds less overhead.

In tunnel mode, the entire original IP packet—including both header and payload—is encrypted and encapsulated within a new IP packet with a new header. This provides a higher level of security and is commonly used in Virtual Private Networks (VPNs), especially for communication between networks or gateways.

22. Differentiate HTTP & HTTPS.

Aspect	**HTTP**	**HTTPS**
Full Form	HyperText Transfer Protocol	HyperText Transfer Protocol Secure
Security	Not secure (no encryption)	Secure (uses SSL/TLS encryption)
Data Transmission	Plain text	Encrypted data
Vulnerability	Prone to eavesdropping and attacks	Protected against interception and tampering
Port Number	80	443
Certificate	Not required	Requires SSL/TLS certificate
Usage	Basic web browsing	Secure transactions (banking, login, e-commerce)

23. How can malware affect your system?

Malware can affect a system in multiple harmful ways, depending on its type and purpose.

It can compromise confidentiality by stealing sensitive information such as passwords, banking details, or personal data. It may also damage integrity by modifying or corrupting files, making data unreliable or unusable. In terms of availability, malware can slow down systems, crash applications, or even make the system completely inaccessible.

Additionally, malware can create backdoors for unauthorized access, monitor user activity, and spread to other systems, increasing the overall impact of the attack.

24. Explain proxy server in details and why we need it?

A proxy server is an intermediary system that acts as a gateway between a user and the internet. When a user makes a request to access a website, the request first goes to the proxy server, which then forwards it to the destination server. The response from the website is sent back to the proxy, which then delivers it to the user.

The need for a proxy server arises from several practical and security considerations. It helps in improving security by hiding the user's IP address, thereby providing anonymity. It can also filter and control access to websites, which is useful in organizations to restrict unwanted content. Additionally, proxy servers can cache frequently accessed data, improving performance and reducing bandwidth usage. They are also used for monitoring and logging user activity for administrative purposes.

25. Explain IP Security Architecture.

IP Security Architecture refers to the framework that defines how IPsec secures communication over IP networks by providing confidentiality, integrity, and authentication.

It is built around key components such as the Authentication Header (AH) and Encapsulating Security Payload (ESP). AH provides data integrity and authentication by ensuring that the data has not been altered and verifying the sender's identity, but it does not provide encryption. ESP, on the other hand, provides confidentiality through encryption, along with optional integrity and authentication.

Another important concept is the Security Association (SA), which defines the parameters for secure communication between parties, including encryption algorithms and keys. These associations are managed through protocols like Internet Key Exchange (IKE), which handles the negotiation and exchange of keys securely.

26. Explain various types of Active & Passive attacks in detail.

A passive attack involves monitoring or listening to communications without altering the data. The attacker's goal is to gather information, such as reading confidential messages or analyzing traffic patterns. Examples include eavesdropping and traffic analysis. These attacks are difficult to detect because they do not modify the system or data.

An active attack, in contrast, involves some form of modification, disruption, or unauthorized interaction with the system. The attacker may alter data, inject malicious information, or disrupt services. Examples include man-in-the-middle attacks, denial-of-service attacks, and data modification. Active attacks are generally easier to detect but more damaging.

27. Define: Malicious software. Explain Virus, Worm, Keylogger, Trojans in detail.

Malicious software, commonly known as malware, refers to any program or code designed to harm, exploit, or disrupt computer systems, networks, or data.

A virus is a type of malware that attaches itself to legitimate files or programs and spreads when those files are executed. It often corrupts or deletes data and can slow down system performance.

A worm is a self-replicating malware that spreads across networks without needing a host file or user action. It consumes system resources and can cause network congestion.

A keylogger is a program that secretly records keystrokes made by a user, capturing sensitive information such as passwords and financial details.

A Trojan is a deceptive malware that appears as legitimate software but performs malicious actions once installed, such as creating backdoors or stealing data.

28. List out various types of cybercrime and explain anyone.

Cybercrime refers to illegal activities carried out using computers, digital devices, or networks, where the computer may be the target, the tool, or both.

There are various types of cybercrime, including hacking, identity theft, phishing, cyber stalking, online fraud, and ransomware attacks.

One important example is phishing. In a phishing attack, the attacker pretends to be a trusted entity—such as a bank or a well-known website—and sends fake emails or messages to trick users into revealing sensitive information like passwords or credit card details. These messages often contain links to fraudulent websites that look identical to legitimate ones.

Phishing is particularly dangerous because it exploits human trust rather than technical vulnerabilities, making it highly effective.

29. Explain Salami attack in detail

A salami attack is a type of cybercrime in which an attacker makes very small, often unnoticed changes or transactions, typically involving financial data, that accumulate over time into a significant loss.

In this attack, the changes are so minor—such as deducting a few paise or cents from many accounts—that individual users are unlikely to notice them. However, when these small amounts are collected from a large number of transactions, the attacker gains a substantial sum.

For example, a malicious programmer in a banking system might alter the software to round off transaction amounts and secretly transfer the fractional differences into their own account. Each transaction appears normal, but collectively they result in financial fraud.

30. Define: Cybercrime. Also discuss needs of Cyber Law.

Cybercrime is defined as any unlawful activity carried out using computers, digital devices, or networks, where the computer can be the target, the tool, or both.

The need for cyber law arises from the rapid growth of internet usage and digital transactions, which traditional legal systems were not originally designed to regulate. Cyber laws provide a structured legal framework to define various cyber offenses and prescribe appropriate punishments. They play a crucial role in protecting the privacy and personal data of individuals, ensuring secure online transactions, and maintaining trust in digital platforms such as banking, e-commerce, and communication systems.

Additionally, cyber laws establish guidelines for digital evidence and legal procedures, making it possible to investigate and prosecute cybercriminals effectively in a rapidly evolving technological landscape.

31. Explain Cyber terrorism.

Cyber terrorism refers to the use of computers, networks, and the internet to carry out terrorist activities that aim to create fear, disruption, or damage to critical systems and infrastructure.

It involves attacks on important sectors such as government systems, banking networks, power grids, transportation systems, or communication networks. The objective is not only to cause technical damage but also to spread panic and instability among people or governments.

Cyber terrorism can take various forms, including hacking into sensitive systems, launching large-scale denial-of-service attacks, spreading malware to disrupt operations, or stealing confidential information related to national security.

For example, an attacker targeting a country's power grid and shutting down electricity supply can create widespread chaos, affecting hospitals, transportation, and daily life, thereby achieving the goals of terrorism through digital means.

32. Explain Web Jacking.

Web jacking refers to a type of cyberattack in which an attacker takes control of a website without the owner's permission, often by hacking into the web server or exploiting vulnerabilities.

Once control is gained, the attacker may modify the website's content, redirect users to malicious or fake websites, or display misleading information. In some cases, the attacker may demand money from the original owner to restore control of the website, making it similar to digital extortion.

Web jacking is dangerous because it can damage the reputation of the website owner, mislead users, and potentially lead to data theft or the spread of malware.

33. Define Phishing, spyware and logic bomb

Phishing, spyware, and logic bombs are distinct types of cyber threats, each targeting systems or users in different ways.

Phishing is a social engineering attack in which an attacker pretends to be a trusted entity, such as a bank or a well-known website, to trick users into revealing sensitive information like passwords or credit card details. It is typically carried out through fake emails, messages, or websites.

Spyware is a type of malicious software that secretly monitors user activity without their knowledge. It can collect information such as browsing habits, login credentials, and personal data, and send it to the attacker.

A logic bomb is a malicious piece of code that remains inactive within a system until a specific condition is met, such as a particular date or event. Once triggered, it executes harmful actions like deleting files or disrupting system operations.

34. Define Web Jacking, Data Diddling, Dos Attack and DDOS Attack

Web jacking, data diddling, DoS attack, and DDoS attack are different forms of cyber threats that target systems and data in various ways.

Web jacking is the unauthorized takeover of a website by an attacker, allowing them to alter content or redirect users to malicious sites.

Data diddling refers to the unauthorized alteration of data before or during its entry into a system. This type of attack is often performed by insiders and can lead to incorrect or manipulated information, especially in financial systems.

A Denial of Service (DoS) attack aims to make a system or network unavailable to its users by overwhelming it with excessive requests or traffic, causing it to crash or slow down significantly.

A Distributed Denial of Service (DDoS) attack is a more advanced form of DoS, where multiple compromised systems (often forming a botnet) are used simultaneously to flood the target, making the attack more powerful and difficult to defend against.

35. Define Cyber bullying, Phishing, spyware and logic bomb

Cyber bullying, phishing, spyware, and logic bombs are different types of cyber threats affecting users and systems.

Cyber bullying refers to the use of digital platforms such as social media, messaging apps, or online forums to harass, threaten, or humiliate individuals. It often involves repeated harmful behavior and can have serious psychological effects on victims.

Phishing is a deceptive technique where attackers impersonate trusted entities to trick users into revealing sensitive information like passwords or banking details, usually through fake emails or websites.

Spyware is malicious software that secretly monitors user activity and collects personal information without consent, often sending it to an attacker.

A logic bomb is a hidden piece of malicious code that activates only when a specific condition is met, such as a certain date or event, and then performs harmful actions like deleting data.

36. Explain Cyber spying and Cyber theft.

Cyber spying and cyber theft are two serious forms of cybercrime, both involving unauthorized access to information but with different objectives.

Cyber spying refers to the act of secretly monitoring or accessing confidential information, typically for intelligence gathering. It is often carried out by individuals, organizations, or even governments to obtain sensitive data such as trade secrets, military information, or strategic plans, without the knowledge of the target.

Cyber theft, on the other hand, involves stealing digital assets or sensitive information for personal or financial gain. This may include stealing credit card details, banking credentials, intellectual property, or personal data.

While cyber spying focuses on surveillance and information gathering, cyber theft is primarily concerned with illegally acquiring valuable data for misuse or profit.

37. Explain Cyber bullying & Cyber stalking.

Cyber bullying and cyber stalking are forms of online harassment, but they differ in nature and intensity.

Cyber bullying involves repeated aggressive or harmful behavior carried out through digital platforms such as social media, messaging apps, or online forums. It includes actions like sending abusive messages, spreading rumors, or publicly humiliating someone. It is often seen among peers, especially teenagers.

Cyber stalking, on the other hand, is more severe and involves continuous monitoring, tracking, or threatening a person online. The attacker may repeatedly send threatening messages, monitor the victim's activities, or attempt to intimidate them over a long period. It can create a sense of fear and danger for the victim.

In essence, cyber bullying is harassment with intent to harm emotionally, while cyber stalking involves persistent surveillance and intimidation, often escalating to serious threats.

38. How we prevent cyber crime with?

Cybercrime can be prevented through a combination of technical measures, legal awareness, and responsible user behavior.

At a fundamental level, using strong and unique passwords, along with multi-factor authentication, significantly reduces the risk of unauthorized access. Regularly updating software and operating systems helps patch security vulnerabilities that attackers might exploit. Installing antivirus and firewall protection adds another layer of defense against malicious activities.

From an organizational and legal perspective, implementing security policies, conducting regular audits, and enforcing cyber laws help deter cybercriminal activities. Education and training further strengthen the ability to detect and respond to threats.

39. Explain article section 66 and section 67 of cyber law

Section 66 and Section 67 are important provisions under the Information Technology Act, 2000 that deal with cyber offenses and penalties.

Section 66 focuses on computer-related offenses such as unauthorized access, data theft, hacking, and damage to computer systems. It applies when someone dishonestly or fraudulently performs acts like accessing a computer without permission, introducing malware, or disrupting services. The punishment may include imprisonment (up to three years), a fine, or both.

Section 67 deals with the publishing or transmission of obscene content in electronic form. This includes sharing inappropriate or sexually explicit material through digital platforms. The penalties are more severe for repeated offenses, involving higher fines and longer imprisonment terms.

Together, these sections aim to regulate misuse of digital technologies and protect users from cyber misconduct.

40. Explain Main in the middle attack with suitable examples

A Man-in-the-Middle (MITM) attack is a type of cyberattack in which an attacker secretly intercepts and possibly alters the communication between two parties who believe they are directly communicating with each other.

In this attack, the attacker positions themselves between the sender and receiver, capturing sensitive information such as login credentials, banking details, or confidential messages. The attacker can also modify the data being transmitted without either party realizing it.

For example, if a user connects to a public Wi-Fi network, an attacker on the same network can intercept the communication between the user and a website. The attacker may steal login information or even alter transactions, such as changing a payment destination.

. 41. Explain ransomware in detail?

Ransomware is a type of malicious software designed to block access to a computer system or encrypt its data until a ransom is paid to the attacker. It is one of the most dangerous forms of cybercrime because it directly targets the availability and integrity of data.

Once a system is infected, ransomware encrypts files using strong cryptographic algorithms, making them inaccessible to the user. The attacker then displays a ransom note demanding payment, often in cryptocurrency, in exchange for a decryption key. Even after payment, there is no guarantee that access will be restored.

Ransomware typically spreads through phishing emails, malicious attachments, infected websites, or software vulnerabilities. It can severely impact individuals, businesses, and even critical infrastructure by causing data loss, financial damage, and operational disruption.

42. Explain article section 66 of cyber law.

Section 66 of the Information Technology Act, 2000 deals with computer-related offenses committed dishonestly or fraudulently. It is essentially an extension of unauthorized access and hacking activities defined under cyber law.

This section applies when a person intentionally performs illegal actions such as accessing a computer system without permission, stealing or copying data, introducing viruses or malware, damaging or disrupting computer systems, or causing loss of information. The key element is the presence of dishonest or fraudulent intent behind the act.

The punishment under Section 66 can include imprisonment for up to three years, a fine, or both, depending on the severity of the offense. This section plays an important role in deterring cybercriminal activities and protecting digital systems from misuse.

43. Explain article section 67 of cyber law.

Section 67 of the Information Technology Act, 2000 deals with the punishment for publishing or transmitting obscene material in electronic form.

This section applies when any person publishes, sends, or transmits content through digital platforms that is obscene or sexually explicit in nature. It includes material shared through websites, emails, social media, or any other electronic communication system.

The objective of this law is to regulate and control the circulation of inappropriate content on the internet and protect users from exposure to harmful material. The punishment under Section 67 includes imprisonment and fines, which are more severe for repeat offenders.

In essence, Section 67 ensures that digital platforms are not misused for distributing obscene content and helps maintain ethical standards in cyberspace.

44. List out any six basic kali Linux commands.

Some commonly used basic commands in Kali Linux are:

ls – used to list files and directories in the current location.
pwd – displays the present working directory.
cd – changes the directory.
mkdir – creates a new directory.
rm – removes files or directories.
ifconfig – displays network interface configuration details.

45. Explain any three basic kali Linux commands.

In Kali Linux, basic commands are used for system navigation and file management.

The **ls** command is used to list all files and directories in the current working directory. It helps users quickly view the contents of a folder.

The **cd** command is used to change the current directory. By specifying a path, users can navigate between different folders in the system.

The **mkdir** command is used to create a new directory. It allows users to organize files by making separate folders for different purposes.

46. Discuss types of Hackers.

White Hat hackers are ethical professionals who work with permission to identify and fix security vulnerabilities in systems. They are often employed by organizations to improve cybersecurity.

Black Hat hackers are malicious individuals who break into systems without permission to steal data, cause damage, or gain financial benefit.

Grey Hat hackers fall between white and black hats. They may access systems without permission but usually do not have malicious intent; however, their actions are still unethical and illegal.

Script Kiddies are inexperienced hackers who use pre-made tools or scripts created by others to perform attacks without deep technical knowledge.

Hacktivists are hackers motivated by political or social causes. They attack systems to promote their beliefs or protest against organizations or governments.

47. Explain Brute force attack.

A brute force attack is a method used by attackers to gain unauthorized access to a system by systematically trying all possible combinations of passwords or encryption keys until the correct one is found.

In this attack, automated tools or software are used to repeatedly attempt login credentials at high speed. The attacker does not rely on intelligence or vulnerabilities but instead depends on trial and error. Simple or short passwords are especially vulnerable to such attacks.

Brute force attacks can be time-consuming and computationally expensive, but they are effective against weak passwords. To protect against them, systems use techniques like account lockout policies, CAPTCHA verification, and enforcing strong password requirements.

48. Describe Hacking, Blackhat, and White hat hacker in short.

Hacking refers to the process of identifying weaknesses in computer systems or networks and exploiting them, either for testing security or for malicious purposes.

A White Hat hacker is an ethical security expert who performs hacking activities with permission to find and fix vulnerabilities in systems. Their goal is to improve security and protect data.

A Black Hat hacker is a malicious individual who illegally breaks into systems to steal data, cause damage, or gain financial benefit. Their actions are harmful and criminal in nature.

49. Explain Zero day attack with example

A zero-day attack refers to a cyberattack that exploits a software vulnerability that is unknown to the software vendor or developers at the time of exploitation. Since the flaw is not yet discovered or patched, attackers can take advantage of it before any security fix is available.

The term “zero-day” indicates that developers have had zero days to fix the vulnerability once it becomes known to attackers. These attacks are highly dangerous because there is no immediate defense available when the exploit is used.

For example, if a hacker discovers a security flaw in a popular web browser and uses it to install malware on users’ systems before the company releases a patch, that would be considered a zero-day attack.

50. What is Port Scanning? Explain any two port scanning techniques.

Port scanning is a technique used to identify open ports and services running on a computer or network. It is commonly used by both security professionals for testing and attackers for finding vulnerabilities in systems.

One common technique is **TCP Connect Scan**, where the scanner attempts to establish a full TCP connection with the target port. If the connection is successful, the port is considered open; otherwise, it is closed. This method is easy to detect but reliable.

Another technique is **SYN Scan (Half-open Scan)**, where the scanner sends a SYN packet but does not complete the full TCP handshake. If a SYN-ACK response is received, the port is open, and the connection is not fully established. This method is faster and more stealthy compared to a full connection scan.

51. Explain RAT.

A Remote Access Tool (RAT) is a type of software that allows a user to remotely control another computer over a network. It is commonly used for legitimate purposes such as technical support and system administration.

However, in the context of cybercrime, a RAT becomes malicious when used without the victim’s knowledge. Attackers install it on a target system to gain full control, enabling them to access files, monitor user activity, capture keystrokes, activate webcams, and execute commands remotely.

RATs are often spread through phishing emails, infected downloads, or malicious attachments. Once installed, they operate in stealth mode, making them difficult to detect. Proper security measures like antivirus software, firewalls, and avoiding suspicious links can help protect against RAT infections.

52. Define: Vulnerability, Threat, Exploit

A vulnerability is a weakness or flaw in a system, software, or network that can be exploited to compromise security. It may exist due to poor design, configuration errors, or outdated software.

A threat is any potential danger that can take advantage of a vulnerability to harm a system or data. It can be natural, accidental, or intentional, such as hackers, malware, or system failures.

An exploit is a method, tool, or piece of code used by an attacker to take advantage of a vulnerability in order to gain unauthorized access, disrupt services, or steal data.

53. How vulnerability scanning operates.

Vulnerability scanning is an automated process used to identify security weaknesses in a system, network, or application. It works by systematically probing the target environment and comparing its configuration, software versions, and services against a database of known vulnerabilities.

The scanner first discovers active devices and open ports within the network. It then examines these systems to detect outdated software, misconfigurations, missing patches, or insecure settings. After analysis, it generates a detailed report listing identified vulnerabilities along with their severity levels and possible remediation steps.

This process helps organizations proactively identify and fix security issues before attackers can exploit them, improving overall system security.

54. Explain Remote Access Tools and how we protect system from RAT?

Remote Access Tools (RATs) are software programs that allow a user to remotely control and access another computer over a network. They are widely used for legitimate purposes such as technical support, system administration, and troubleshooting.

However, in cybercrime, RATs are often used maliciously without the user's knowledge. Attackers use them to gain unauthorized control over a system, enabling activities like stealing files, monitoring user actions, recording keystrokes, activating webcams, and executing commands remotely.

To protect systems from RAT attacks, several security measures are important. Installing and regularly updating antivirus and anti-malware software helps detect and remove suspicious programs. Firewalls should be enabled to block unauthorized remote connections. Users should avoid downloading files from unknown sources or clicking on suspicious email attachments and links. Keeping operating systems and software updated also reduces vulnerabilities that RATs may exploit.

55. Explain the hacking process in detail.

The hacking process refers to a series of steps followed by attackers to gain unauthorized access to a system or network. It generally begins with information gathering, where the attacker collects details about the target such as IP addresses, domain information, and system configuration.

The next step is scanning, where tools are used to identify open ports, active services, and possible vulnerabilities in the system. After this, the attacker moves to gaining access by exploiting identified weaknesses using techniques such as password cracking, malware injection, or exploiting software flaws.

Once access is obtained, the attacker may maintain access by installing backdoors or Remote Access Tools, allowing continued entry into the system even if the original vulnerability is fixed. The final

stage is covering tracks, where the attacker tries to erase logs and hide evidence of the intrusion to avoid detection.

This structured process helps attackers systematically identify and exploit weaknesses in a target system.

56. Explain Five Steps of Hacking.

The hacking process typically consists of five main steps followed by attackers to gain unauthorized access to a system.

The first step is **reconnaissance (information gathering)**, where the attacker collects as much information as possible about the target, such as IP addresses, domain details, user information, and system architecture.

The second step is **scanning**, where tools are used to identify open ports, active services, and possible vulnerabilities in the target system.

The third step is **gaining access**, where the attacker exploits identified vulnerabilities using techniques like password cracking, phishing, or malware injection to enter the system.

The fourth step is **maintaining access**, where the attacker installs backdoors or malicious tools to ensure continued access to the system even after the initial breach is detected or fixed.

The fifth and final step is **covering tracks**, where the attacker attempts to hide their activities by deleting logs, clearing evidence, and masking their presence to avoid detection.

57. Explain any three basic commands of kali Linux with suitable example.

In Kali Linux, several basic commands are used for navigation and system control.

The **ls** command is used to list files and directories in the current working directory. For example, typing ls displays all files present in a folder.

The **cd** command is used to change the directory. For example, cd Desktop moves the user into the Desktop directory.

The **mkdir** command is used to create a new directory. For example, mkdir test creates a folder named “test” in the current location.

58. Write Locards's principal and explain how it is related to cybercrime?

Locard’s Principle of Exchange states that whenever two objects come into contact, there is always a transfer of material between them. In simple terms, “every contact leaves a trace.”

In cybercrime and digital forensics, this principle applies because any interaction with a computer system leaves behind digital traces. When a cybercriminal accesses a system, they inevitably leave evidence such as log files, IP addresses, timestamps, browser history, or temporary files.

This principle is highly important in cyber investigations because forensic experts rely on these digital traces to reconstruct events, identify attackers, and understand how a cybercrime was committed. Even if an attacker tries to erase evidence, some form of residual data usually remains.

59. List the branches of Digital Forensics.

Computer forensics focuses on analyzing data from computers, laptops, and storage devices to find evidence of cyber activities.

Network forensics deals with monitoring and analyzing network traffic to detect intrusions and attacks.

Mobile device forensics involves extracting and examining data from smartphones, tablets, and other portable devices.

Database forensics is concerned with investigating databases to identify unauthorized access, data manipulation, or theft.

Malware forensics focuses on analyzing malicious software to understand its behavior, origin, and impact.

60. List critical steps in preserving digital evidence.

The first step is **identification**, where potential sources of digital evidence such as computers, mobile devices, or storage media are located.
The second step is **securing the scene**, ensuring that the system is not tampered with or accessed by unauthorized individuals.
The third step is **collection**, where evidence is carefully extracted using forensically sound methods without altering the original data.
The fourth step is **imaging**, where an exact bit-by-bit copy of the storage device is created for analysis.
The fifth step is **preserving integrity**, where hash values are used to verify that the evidence has not been modified.
The final step is **maintaining chain of custody**, which records every person who handled the evidence along with time and purpose.

61. Explain four stage of malware analysis in short.

Malware analysis is the process of examining malicious software to understand its behavior, functionality, and impact.

The first stage is **static analysis**, where the malware is examined without executing it. Analysts study file properties, code structure, strings, and signatures to gather initial information.

The second stage is **dynamic analysis**, where the malware is executed in a controlled environment such as a sandbox to observe its real-time behavior, including file changes, network activity, and system modifications.

The third stage is **code analysis (reverse engineering)**, where the internal code of the malware is deeply examined to understand its logic, algorithms, and techniques used for infection or concealment.

The fourth stage is **reporting and documentation**, where all findings are recorded, including behavior patterns, risks, and possible countermeasures to help in detection and prevention.

62. Explain Malware forensics.

Malware forensics is a branch of digital forensics that focuses on identifying, analyzing, and understanding malicious software and its behavior in a system or network. The main objective is to determine how the malware entered the system, what damage it caused, and how to prevent future attacks.

The process involves collecting infected files, memory dumps, and system logs for detailed examination. Analysts study the malware's code, execution patterns, and communication methods with external servers. Techniques such as static analysis and dynamic analysis are commonly used to understand its structure and behavior.

Malware forensics also helps in identifying the source of the attack and the techniques used, such as phishing emails, infected downloads, or exploit kits. The findings are used to develop security measures, create detection signatures, and improve system defenses against similar threats.

63. What is software forensics? How it is contributing in cybercrime?

Software forensics is a branch of digital forensics that focuses on analyzing software, source code, and executable programs to determine their origin, functionality, and any malicious modifications. It helps investigators understand how a program behaves and whether it has been used for illegal or harmful purposes.

In cybercrime investigations, software forensics plays a crucial role in detecting tampered or pirated software, identifying malicious code embedded in applications, and tracing unauthorized modifications. It helps determine whether software was intentionally altered to perform harmful actions such as data theft, system disruption, or unauthorized access.

It also contributes by supporting legal cases through evidence extraction, helping identify developers or attackers behind malicious software, and improving overall cybersecurity by revealing vulnerabilities in programs.

64. Discuss Locard's Principle of Exchange in Digital Forensics.

Locard’s Principle of Exchange states that whenever two objects come into contact, there is always an exchange of material between them, meaning every contact leaves a trace.

In digital forensics, this principle applies to every interaction with a computer system or network. When a user or attacker accesses a system, they inevitably leave behind digital traces such as log files, IP addresses, timestamps, registry changes, browser history, or temporary files.

Even if an attacker tries to delete evidence, traces often remain in system memory, backup files, or hidden logs. Forensic experts use these traces to reconstruct events, identify the attacker, and understand the sequence of actions performed during a cybercrime.

65. How does network forensic functions?

Network forensics is a branch of digital forensics that focuses on monitoring, capturing, and analyzing network traffic to detect and investigate security incidents.

It functions by continuously collecting data packets that move across a network using tools like packet sniffers and intrusion detection systems. These packets are then stored for analysis, allowing investigators to examine communication between devices.

During analysis, experts reconstruct network sessions to understand how data was transmitted, identify unauthorized access attempts, and detect malicious activities such as data exfiltration or attacks. Logs from routers, firewalls, and servers are also examined to trace the source and destination of traffic.

66. Explain why CCTV plays an important role as evidence in digital forensics investigations.

Closed-Circuit Television (CCTV) plays a crucial role in digital forensic investigations because it provides visual evidence of events as they actually occurred. It helps investigators verify activities, identify suspects, and reconstruct crime scenes with accuracy.

CCTV footage can capture real-time actions such as unauthorized access, theft, or suspicious behavior, which can be used to support other digital evidence like logs or system records. It also helps establish timelines by showing exactly when an incident took place.

In many cases, CCTV recordings serve as strong legal evidence in court because they provide direct visual proof of an event. Additionally, modern CCTV systems often include timestamps and metadata, which further strengthen their reliability in forensic analysis.

67. Explain Mobile Device Forensics in detail

Mobile device forensics is a branch of digital forensics that focuses on recovering and analyzing data from mobile devices such as smartphones, tablets, and GPS units. It is used in cybercrime investigations to extract evidence like call logs, messages, emails, photos, videos, application data, and location history.

The process begins with **acquisition**, where data is carefully extracted from the device without altering the original information. This may involve logical acquisition (accessing visible data) or physical acquisition (creating a bit-by-bit copy of the device memory).

The next stage is **examination and analysis**, where investigators study the extracted data to identify relevant evidence, recover deleted files, and analyze communication patterns or user activity.

Finally, in the **reporting stage**, all findings are documented in a structured format to be presented as legal evidence in court.

68. Explain in detail: Drive imaging, Chain of custody and hash values

Drive imaging is the process of creating an exact bit-by-bit copy of a storage device such as a hard disk or USB drive. This copy, called a forensic image, includes all files, deleted data, and hidden information. Investigators use the image for analysis so that the original evidence remains unchanged.

Chain of custody refers to the documented process that tracks the movement and handling of digital evidence from the time it is collected until it is presented in court. It records every individual who accessed the evidence, along with the time, date, and purpose of access. This ensures the integrity and authenticity of the evidence is maintained.

Hash values are unique digital fingerprints generated using cryptographic algorithms like MD5 or SHA. They are used to verify that data has not been altered. If the hash value of the original evidence matches the hash value of the forensic image, it confirms that the data remains unchanged.

69. Explain digital forensic investigation process

It begins with **identification**, where potential sources of evidence such as computers, mobile devices, or networks are recognized.

The next step is **preservation**, ensuring that the original evidence is protected from any modification or damage.

The third stage is **collection**, where data is carefully extracted using forensic tools while maintaining integrity.

After collection, the **examination and analysis** phase is performed, where investigators study the data to find relevant evidence, recover deleted files, and reconstruct events.

The final stage is **reporting and presentation**, where all findings are documented clearly and presented in court or to relevant authorities as evidence.

70. List the critical steps in preserving Digital Evidence.

The first step is **identification**, where potential sources of evidence such as computers, mobile devices, storage media, or network logs are located.
The second step is **isolation**, where the device is disconnected from networks to prevent remote access or tampering.
The third step is **securing the scene**, ensuring that unauthorized persons do not alter or access the evidence.
The fourth step is **forensic acquisition**, where a bit-by-bit copy (image) of the original data is created.
The fifth step is **verification using hash values**, which ensures that the copied evidence matches the original without any modification.
The final step is **chain of custody maintenance**, where every handling of the evidence is documented with details of time, person, and purpose.

71. Explain phases of Digital forensic investigation.

The first phase is **identification**, where investigators determine potential sources of digital evidence such as computers, mobile devices, storage media, or network systems involved in the incident.

The second phase is **preservation**, where the original evidence is protected from alteration or damage. This includes isolating devices from networks and ensuring no changes occur to the data.

The third phase is **collection**, where digital evidence is carefully acquired using forensic tools. A forensic image or exact copy of the data is created to preserve the original evidence.

The fourth phase is **examination and analysis**, where investigators study the collected data to extract relevant information, recover deleted files, and reconstruct events related to the cybercrime.

The final phase is **reporting and presentation**, where all findings are documented in a structured report and presented in court or to authorities as legal evidence.